

ОКБ-256 · САН-MATEO — ДОКУМЕНТ
EXPERIMENTAL DESIGN BUREAU 256, 256-00
SAN MATEO

ИЗДЕЛИЕ 256 – PROGRAM BASELINE

MY FIRST JET ENGINE ·
LINEAR VALVELESS PULSEJET

REV. H – ЗАМОРОЖЕНО / FROZEN

CHIEF DESIGNER

Alex

DEPUTY CHIEF DESIGNER

Mr. Napkin

CONSULTING ENGINEERS

Claude, GPT

Duration **1–3 months**

Budget ~\$**650–850** all-in

Articles **256B** primary · **256B** contingency

Welding **outsourced**

Single-builder garage bureau; this is the Chief Designer's first hardware program (software engineer by trade — the plan accounts for it). Tools: Bambu

*Lab P1S, Makera Carvera Air **standalone** (4th-axis and laser modules: capital request denied without prejudice — refile under 256B when a drawing contains a cylinder), full CAD (Fusion 360 / SolidWorks / NX) and CAE (Ansys / Simcenter) access. No welding capability (outsourced). This document supersedes Rev. G, which declared there would be no Rev. H; Rev. H exists by direct order of the Chief Designer. There is no Rev. I. The next document is drawing 256-01: a rectangle with holes in it.*

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§ 1 OBJECTIVE

Build one instrumented test facility and one propulsion article; run a disciplined test campaign with a pre-test prediction, stated uncertainty, and honest correlation; present the result as a deadpan Soviet experimental-program homage. Audience: aerospace engineers and executives (NASA, Airbus, SpaceX) and skilled hands-on traditionalists. Desired reaction: "*why does this exist*" → "*wait, this is real engineering.*" Governing aesthetic law: **crude design, precise execution — the gap is the joke.** Secondary objective: the program doubles as the Chief Designer's introduction-to-CNC curriculum — real machined parts with zero-stakes fallbacks. Tertiary objective, added this revision: the finished article must read as *gorgeous to non-engineers* using only approved cold-side and material means — beauty is achieved through metallurgy, mixture, and lighting, never through structural risk.

§ 2 PROGRAM ARCHITECTURE

– TWO ARTICLES, ONE GATE

Common infrastructure begins immediately; the article is selected at a formal gate, so the open land question costs zero schedule.

Linear valveless pulsejet,
Lockwood/SNECMA-family geometry.
*256B uses a Western valveless
architecture; the Soviet treatment is a
design-bureau homage, not a claim of
engine lineage.* **Hot section in 304
stainless, polished before first fire** —
the engine's own heat then lays down
permanent straw-gold-purple-blue
temper coloration graded along the tube
(the single highest-yield gorgeousness
decision in the program; ask the shop
about TIG for stainless when booking —
mild steel remains the cost/availability
fallback). Propane, $\sim 1\text{--}2$ kgf mean
thrust, $\sim 100\text{--}250$ Hz, ~ 140 dB, hot
section glows in daylight. Requires
authorized drive-out test land.

Standing prohibitions: no garage,
driveway, or municipal firing, ever, for
any duration, under any rationalization
— 140 dB is indifferent to intent. And
no transparent combustion section on

Mk I: the tube wall is the chamber, so "see-through housing" means a quartz combustion vessel — a real but genuinely difficult lab-engineering problem (thermal-mismatch sealing on a structure vibrating at 200 Hz; failure mode is hot glass on a running engine). It is filed with its own designation, **Изделие 256С (стеклянный)**, for a future plan, on a second engine. A transparent *outer* shroud is equally denied: plastic melts in the radiant field and there is no cool place to put a window on a machine that is entirely hot section. Mk I is opaque by design and glows anyway.

ARTICLE 256Θ · CONTINGENCY, HELD
WITHOUT SHAME

~70 mm commercial EDF core with
printed inlet/duct/nozzle variant
families and machined interfaces.
Thrust-per-kW campaign, garage-
testable nightly. LiPo charge/storage
discipline is this article's principal fire
risk and gets a written protocol. 256Θ is
also the designated **exit door**: if, at any
point, the propane sections of this
program read as dread rather than
respect, the electric article delivers
~80% of the program at RC-hobby
hazard levels, and taking that door is a
program decision, not a defeat.

THE GATE — END OF WEEK 3 • CALLS
BEGIN WEEK 1

explicit confirmation from the landowner or club AND the authority having jurisdiction that this specific propane-fed static propulsion test is permissible under current fire restrictions. "It's BLM land" establishes nothing; a campfire permit is not evidence. **Site authorization is the dominant critical path — above welding.**

Site A (strongly preferred): an AMA club within day-trip range that already hosts turbine/pulsejet operation. Site B: private rural land with written permission and an AHJ check. From San Mateo: Central Valley and East Bay hinterlands, 1.5–3 hours; sessions are planned expeditions. Firm authorization → 256B; otherwise → 256D without mourning, 256B held in reserve. Both bolt to the same stand. **The contingency does not get a contingency:** if welding falls through —

second shop, then 256Э. The threaded-pipe improvisation remains deleted.



Photo 1. Reference imagery, foreign bureaus: Lockwood-Hiller-family valveless article in polished stainless on a test stand – the geometry family and finish 256B targets.



Photo 2. Night run: blue intake flame, tailpipe at orange heat.
(pulsejetengines.com)



Photo 3. Small article, outdoor static run at dusk. (Beck-Technologies)



Short Lady original prototype test run.
Photo Copyright 2004 Steve Bukowsky

Photo 4. "Short Lady"
prototype at full heat
– why the polish
matters. (© 2004 Steve
Bukowsky)

Reference photographs retained from program
research; watermarks preserved. These are other
builders' engines – the target class of machine,
not the article itself.

§ 3

WORKSTREAM

FACILITY

WEEKS 1–
4 · ARTICLE–
INDEPENDENT

Thrust stand: fixed base → load cell behind a
radiant heat shield → sliding engine carriage
→ article. One fixed mount, one sliding
mount; slider travel sized by Analysis C with
margin. **Load cell by requirement, not
preselection:** capacity chosen after
establishing maximum credible stand load
with overload margin; mechanical travel stops

and overload protection independent of the sensor; purchase deferred until stand geometry is fixed. **Fastener retention by engineering:** proper torque and preload first; locking features by location and temperature; no nylocs in heat-soak zones (this now explicitly includes the shroud region, §7); safety wire only where hardware and failure mode justify it.

Carvera apprenticeship ladder — five real parts in difficulty order, each with a catalog fallback (~\$30 insurance kit). All flat 2.5D on the standalone 3-axis machine — the ladder contains zero rotary parts by design. When asked "what did you machine," the answer is a parts list, not a sign.

256-01	Load-cell plate
256-02	Engine support bracket (the program's L-bracket)
256-03	Sliding expansion support (first slot)

256-04

Control panel (pockets, holes, engraving, chamfers)

256-05

Brass data plate (the reward)

P1S role: welding fixtures, V-blocks, drill templates, interface fit-check sections, knobs, blower adapter, crate stencils, and (new) the cutaway teaching model, §7. No full 1:1 mockup. All plastics ≥ 30 cm from hot metal.

Electrical doctrine, non-negotiable: safety chain entirely electromechanical — ПУСК master-arm (enormous, guarded, functional: physical series contact in the low-voltage supply) → held **dead-man** → АВАРИЙНЫЙ СТОП breaking everything → NC fuel solenoid (256Б) or ESC arm/throttle-cut (256Э). The Arduino (HX711, MAX31855, V/I) reads everything and gates **nothing**; behavior must be identical with the microcontroller unplugged mid-run. **Software observes; copper decides.** No PCB, no Pi, no state-machine theater. Perfboard is period-correct — and now deliberately *beautiful* perfboard, per §7's clear-bay option.

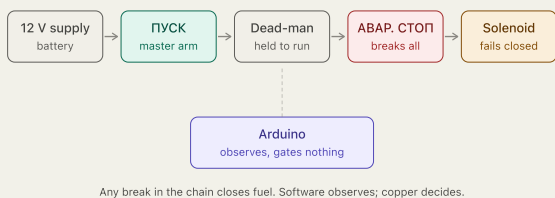


Рис. 1. Hardwired safety chain. Every contact in series is physical; the microcontroller observes and gates nothing.

§ 4 WORKSTREAM ARTICLE

256Б: proven published valveless planset, dimensions locked week 1, zero geometric freelancing — the engine is a tuned resonator. 304 stainless exhaust tube and bought reducer cones (§2), ~6 welds, outsourced to a shop **booked in week 1**; builder arrives with printed fixtures holding final alignment and a weld drawing — buying minutes of arc time, not a builder. Shop may also cut tubes to length in the same visit (see §6, grinder demotion).

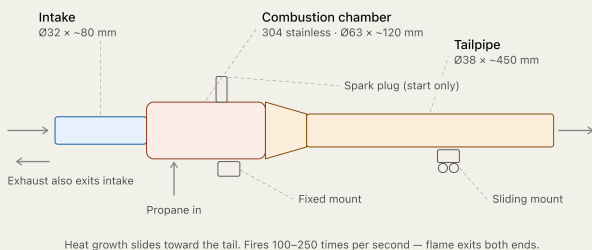


Рис. 2. Article 256B cross-section, Rev. H. 304 stainless throughout the hot section; one fixed mount, one sliding mount for thermal growth; exhaust pulses exit both ends.

Fuel train: cylinder valve → LP-rated regulator → needle valve → NC solenoid → injector (drilled copper baseline; brass orifice as Carvera flair). Every component rated for LP-gas service; hose short, heat-routed; final parts list reviewed at a genuine propane supplier's counter before first fuel. **Mixture configurations, logged:** lean/best-thrust for the test campaign, and **режим «ПАРАД»** — a documented rich setting (needle valve past best thrust) producing a long pulsing exhaust plume and visible intake flame for demonstrations; parade runs carry a deepened downstream exclusion zone and appear as a configuration in the test table like everything

else. **Ignition:** automotive coil, continuous spark during start; piezo backup; spare plugs. **Start air:** leaf blower with sacrificial printed adapter — the contrast is preserved by decree.

256Θ: reputable 70 mm EDF and matched ESC; arming plug on the hardwired chain; variant families = nozzle exit area sweep ($\sim 85\text{--}100\%$ of fan area) and inlet lip geometry, three articles each; optional momentum-source CFD credited **only for predicted deltas**, never absolutes.

§ 5 WORKSTREAM TEST

Four calculations, all hardware-affecting: **(A)** first-order acoustic estimate with open-end corrections, published pre-test as a predicted **band** with stated sound-speed assumptions; **(B)** one-hour stand modal sanity check (eigenfrequency run or cold tap test) confirming no structural mode in the forcing band — after which the report may lawfully state *structural design methodology: sufficiently*

Soviet amounts of steel; **(C)** thermal growth sizing the slider; **(D)** a ten-minute temperature estimate for any 6061 mounted near the shroud region (§7) — aluminum softens by $\sim 200\text{--}300^\circ\text{C}$, so the rings either sit at demonstrably cool stations, clear the calculation, or become brushed stainless. Philosophy: **calculate when the answer affects hardware; never simulate because Ansys has buttons.**

Baseline measurement set (three, and no more): quasi-static time-averaged thrust (HX711 at 10–80 S/s is *explicitly not a dynamic force transducer*; known-weight calibration before **and after** every session; uncertainty and repeatability reported). Skin temperature at one consistent location, inconel-sheathed K-type $\times 2$ (the first is a consumable) — **read in both shroud-on and shroud-off configurations once T-03 exists**, since a radiation shield reflects heat back at the wall. Dominant frequency via phone spectrogram at distance — frequency only, never SPL; clipping honestly reported. Stretch (L4): fuel mass via luggage scale, contingent

on demonstrated resolution; test duration is set by the article and site, never by a \$15 scale. 256 Θ analog: thrust-per-kW.

Graduated success:

- L1 Ten seconds self-sustained / first instrumented run — the project fundamentally works.
- L2 Repeatable starts.
- L3 Full baseline dataset.
- L4 Controlled campaign, one variable at a time, runs logged 256-001 onward.

Crown deliverable, one sentence: "Predicted 190–215 Hz; measured 178; –8% from band edge; attributed to mean-gas-temperature overestimate."

§ 6 TEST OPERATIONS, SAFETY 2565 & BEGINNER DOCTRINE

Control hierarchy: **distance** → **remote operation** → **exposure time** → **PPE**.

Established exclusion zone (deepened downstream for ПАРАД runs); operator at maximum dead-man lead; minimum run durations; double hearing protection as the last layer, never the plan — one unprotected close exposure can be permanent.

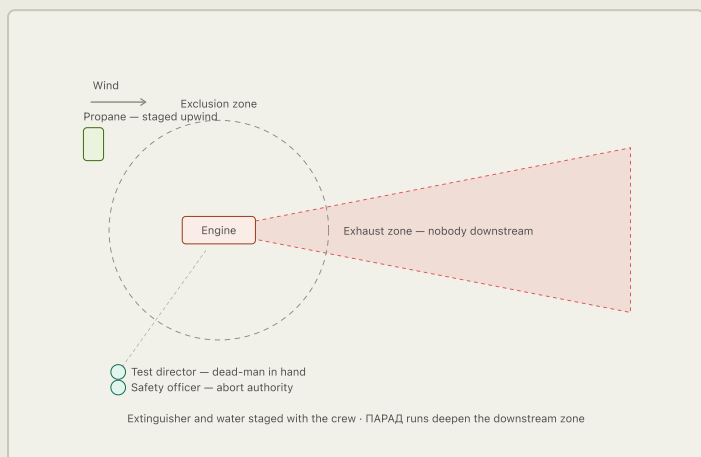


Рис. 3. Hot-fire range layout, top view. Exclusion zone deepened downstream for ПАРАД configuration runs; water staged beyond the extinguisher; nobody downwind.

Minimum hot-fire crew: two. TEST DIRECTOR (dead-man in hand) and SAFETY OFFICER — briefed on shutdown, fuel isolation, extinguisher/water stations,

exclusion watch, independent abort authority. The killer scenario is mundane: one person cannot fight a grass ember and close a cylinder valve simultaneously. Cold work remains single-operator.

Weather gate: red-flag warnings or burn restrictions scrub the session, no debate; water staged beyond the extinguisher; bare dirt/gravel; nobody downwind. **Hot-work separation:** no spark-producing work in the fuel/test area, ever; tuning loop = shut down → isolate fuel → cool → cold-work area → modify → return. **Cold commissioning before the article travels:** soapy-water leak check (bottle outdoors), spark bench test, load-cell calibration, dead-man/E-stop verification — the article arrives at the range with exactly one untested behavior. **Post-run:** no-touch cooldown; gloves after; metal stays branding-hot long after it stops glowing — the classic beginner injury is touching it twenty minutes early.

Beginner doctrine (new, binding — the Chief Designer has built only plastic until

now): The plan itself is the primary safety equipment; it protects only when followed *precisely when it is annoying*. Software habits — cheap, reversible, rapid iteration — do not transfer; improvisation under frustration is where hardware hurts people, and the field moment of "*let me just try one thing*" is hereby designated a forbidden sentence alongside its ancestor about 12 mm. **Grinder demotion:** the angle grinder is the highest injury-density tool in the program and is *deferred* — the weld/muffler shop cuts tubes to length in the booking visit; if a grinder is later acquired, first hour on clamped scrap, face shield, never freehand, never at the field. **Calibration afternoon:** before fabrication month ends, one session absorbing what *normal* looks and sounds like — a maker-space metal-shop intro, or ten minutes of the weld-shop's time watching you drill and deburr. Not buying skills; buying calibration. **Exit door standing:** §2's 256D clause may be exercised at any time without ceremony.

§ 7 **WORKSTREAM** **THEATER**

FUNDED ONLY
AFTER SUCCESSFUL
COMMISSIONING

Joke-placement doctrine unchanged — one joke, one home, descending volume: crate announced in English; artifact discovered in straight-faced Russian; report contains no joke at all, which is the joke.

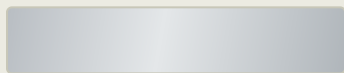
Material palette (new, program-wide): material contrast reads as "expensive" to everyone; monochrome reads only to engineers. The engine itself stays honest bare metal — welds, plumbing, load path inspectable; no paint on the article, ever — its heat coloration is the finish, applied by the engine itself.



RAW STAINLESS, HEAT-
RAINBOWED



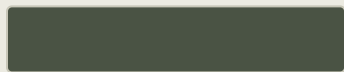
BRUSHED DARK PERFORATED
STEEL



BRIGHT MACHINED 6061



BRASS PLATE



WRINKLE GREY-GREEN
(STAND & PANEL ONLY)

DARK BRAIDED SLEEVING,
ORDERLY ROUTING

Theater options, executed in this order if time runs short (cheapest wow first):

- T-01** **Polish before first fire** (free; enables the temper rainbow).
- T-02** **ПАРАД + dusk filming** (free; the 45-second deliverable is shot in the twenty minutes of civil dusk — glowing translucent-orange steel, twin-ended flame, visible hardware, one phone on slow-mo to resolve individual pulses).

T-03

**ПЕРФОРИРОВАННЫЙ
ТЕПЛОВОЙ КОЖУХ (perforated
thermal shroud):** CNC ring →
perforated cylinder → glowing
chamber visible through holes →
CNC ring — the closest legal
approach to the transparent-turbojet
look. **Approved with thermal
conditions:** $\geq 40\text{--}50\%$ open area,
15–20 mm radial standoff, partial
coverage; it is a logged *configuration*
(it reflects heat back at the wall —
see §5 skin-temp requirement); ring
material per Analysis D; heat-soak
fastener rules apply. Removable; last
in the queue because it alone must
negotiate with thermodynamics.

T-04

Clear cold-side electronics bay:
polycarbonate-lidded enclosure at
the stand's *cool* end — labeled
terminal blocks, relays, Arduino,
braided wiring, deliberately
beautiful. The transparent-machine
itch, scratched at 25 °C.

256-06 · УЧЕБНЫЙ ПЛАКАТ

teaching kit: engraved/printed Soviet-style training placard (half-section, four numbered cycle phases, bilingual captions) plus a P1S-printed cutaway display model.

Official phrasebook entry: *"This one's the cutaway. This one's the one you're not allowed to stand near."*

Crate: hand-built plywood, cleats, rope handles (~\$40–60; explicitly not a CNC job), knife-cut foam cavity, P1S stencils, spray paint. Marking, customs-manifest register:

CONTENTS: ONE (1) PROPULSION DEVICE / FUNCTION: EXTREMELY LOUD NOISE, / MODEST AMOUNT OF THRUST · THIS SIDE UP · small ИЗДЕЛИЕ 256 as the only Cyrillic breadcrumb. Ammo crate acceptable. Month 3 only — wood before fire is the wrong order of operations.

Artifact plate: brass, ИЗДЕЛИЕ 256 · ОПЫТНЫЙ ОБРАЗЕЦ № 001, hand-stamped serial, small ОТК ПРИНЯТО.

Humorless. No modern-Russia iconography, no

ideology posters, no announced jokes on hardware.

Report: one page, institute format, masthead as above, honest error bars, and the two-requirement verification matrix — **R-001** *extremely loud noise* — VERIFIED BY TEST; **R-002** *modest amount of thrust* — VERIFIED BY TEST. *All requirements met. Program declared a complete success.* Verbal deliverable: "**It met all requirements.**" Final deliverables: the dusk video, the heat-rainbowed artifact, the report, the crate, the teaching kit.

§ 8 SCHEDULE & BUDGET

PERIOD	WORK
Week 1	Immediately: site-authorization calls (dominant critical path); weld shop booked (confirm TIG/stainless; request tube cutting); single bulk procurement order — instrumentation and electrical chain, tool tier (bench vise, drill + step bit, soldering kit, PPE including extinguisher; grinder deferred per §6), fallback hardware kit, planset + stainless or EDF pending gate.
Weeks 1–2	CAD assembly; machine 256-01 and 256-02; calibration afternoon scheduled.
Week 3	THE GATE — article locked.
Weeks 3–6	Article fabrication (welding errand or EDF integration); stand complete; cold commissioning; panel wiring.
Weeks 6–9	First fire / first run; success levels 1–3.
Weeks 9–12	L4 campaign; report; then and only then theater T-01 → T-05 in order, plus the crate. Slack absorbs shipping, weather scrubs, and scrapped parts.

BUDGET LINE	ESTIMATE
Article — 256B (stainless delta) or 256C	\$280–320 / \$350–450

BUDGET LINE	ESTIMATE
Stand + instrumentation	~\$100
Tools + PPE (grinder deferred; all tools permanent acquisitions)	\$250–300
Theater + crate + shroud	\$100–125
Program total, all-in	~\$650–850

§ 9 EXCLUDED BY DECREE

Turbine cores, compressor wheels, liquid fuel, custom fuel pumps, custom PCB, Raspberry Pi, FADEC theater, reacting-flow CFD, acoustic CFD, lathe acquisition, 4th-axis and laser modules (denied without prejudice), **transparent combustion sections and quartz windows (filed as Изделие 256С, стеклянный, for a future plan)**, transparent outer shrouds in the radiant field, additional machinery of any kind, garage hot-fire under any rationalization, the threaded-pipe engine, field improvisation of every species, and every attractive idea that arrives after this sentence — each filed as **Изделие 256В** for the next five-year plan.

§ 10 DEFINITION OF FINISHED

A real propulsion article, fabricated by the Chief Designer on his first hardware program, runs repeatably on a stand the Chief Designer machined; produces mean thrust, temperature, and frequency data with stated

uncertainty; matches — or honestly misses, with attributed error — a pre-test prediction; is gorgeous at dusk to any human alive; and only then receives its uniform, its humorless nameplate, its two-requirement report, its teaching kit, and its loud-mouthed crate. The story afterward: *"I wanted a small fabrication project, so naturally I established a Soviet propulsion design bureau in my garage."*

**ИЗДЕЛИЕ 256 · ОПЫТНЫЙ
ОБРАЗЕЦ № 001**

ОТК ПРИНЯТО · ОКБ-256 · САН-МАТЕО

PROGRAM BASELINE REV. H — FROZEN. THERE IS
NO REV. I.